

Adamson T. dela Cruz

Principal AI, Cloud, IoT & Platform Engineer

AWS Certified Solutions Architect – Professional
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New Zealand citizen · English and Filipino

PROFESSIONAL PROFILE

Principal AI, cloud, IoT, and platform engineer with 25+ years of experience delivering production software across embedded devices, real-time telemetry, cloud-native platforms, data pipelines, and enterprise systems. Combines hands-on full-stack engineering with deep experience in AWS, distributed and event-driven architecture, DevOps, platform automation, networking, and IoT.

Experienced across telecom, logistics, manufacturing, finance, government, healthcare, insurance, aviation, construction, education, and digital media. Known for turning complex operational needs into practical, scalable systems and for working effectively with stakeholders, engineering teams, suppliers, and end users.

Now focused on engineering practical AI-enabled systems and AI-augmented delivery workflows. Works actively with OpenAI Codex, Claude, OpenClaw, Qwen, DeepSeek, and Ollama, with particular interest in local LLM deployment, agentic workflows, intelligent automation, and the integration of AI with cloud, IoT, and edge environments.

CORE EXPERTISE

AI engineering: AI-enabled systems, AI-assisted development, local LLM deployment, agentic workflows, intelligent automation, edge AI, OpenAI Codex, Claude, OpenClaw, Qwen, DeepSeek, Ollama

Cloud & architecture: AWS, Azure, serverless, microservices, event-driven systems, distributed systems, solution and enterprise architecture

IoT & edge: AWS IoT Core, MQTT, LoRaWAN, NB-IoT, Cat-M1, 5G, Wi-Fi, embedded systems, gateways, telemetry, RF troubleshooting

Data & integration: AWS Glue, Athena, data lakes, Medallion Architecture, Kafka, Kinesis, DynamoDB, PostgreSQL, SQL Server, MongoDB, Redis

Platform, reliability & security: Kubernetes, Docker, Terraform, CloudFormation, AWS CDK, Ansible, Packer, Chef, Jenkins/CloudBees, Azure DevOps, CI/CD, Linux, observability, IAM, OWASP remediation

Software engineering: Python, TypeScript/Node.js, Go, C#/.NET, Java, C/C++, Bash, React, Vue, Angular

SELECTED CAREER HIGHLIGHTS

- Built AWS serverless services and Bronze/Silver/Gold data pipelines for an industrial sorting and optimisation platform.
- Designed connected platforms spanning AWS IoT, MQTT, LoRaWAN, NB-IoT, telematics, Wi-Fi, and 5G video.

- Helped establish Westpac's centrally managed, configuration-as-code CI/CD platform using CloudBees, Kubernetes, Terraform, Chef, and security automation.
- Founded **IoT Ninja** and designed its cloud platform, reusable applications, private LoRaWAN infrastructure, gateways, and device integrations.

PROFESSIONAL EXPERIENCE

Fullstack Solutions Limited — Director & Consulting Engineer

2022–Present · New Zealand / Remote

Provides senior engineering, architecture, cloud, IoT, and platform automation services to clients in New Zealand and overseas.

SELECTED CLIENT ENGAGEMENTS

Mindhive Global — Senior Full-Stack Engineer

Dec 2024–Jun 2026 · Part-time, remote

Led AWS serverless and data-lake engineering for the JBS Sorting & Optimizer platform supporting tannery inventory and pallet-selection operations.

- Designed and built AWS cloud infrastructure and TypeScript backend services.
- Created AWS Glue pipelines using Bronze, Silver, and Gold data layers.
- Automated build, test, and multi-environment deployment workflows.

Technology: AWS Glue, Lambda, Athena, Redis, PostgreSQL, TypeScript, React, Bash, Linux, CI/CD, Medallion Architecture

ESS — AWS IoT Cloud Developer

Jun 2024–Dec 2025 · Part-time, remote · Dallas, USA

Provides IoT and cloud engineering for a next-generation vehicle safety platform.

- Designed AWS infrastructure and serverless backend services in TypeScript.
- Integrated connected devices through AWS IoT Core and MQTT.
- Built automated test and deployment workflows for multiple environments.

Technology: AWS IoT Core, AWS serverless services, MQTT, TypeScript, Vitest, Bash, Linux, Make

Westpac — Senior Platform Engineer

Sep 2022–Jun 2024 · Auckland, New Zealand

Supported a centralised CI/CD platform for the organisation using an on-premises CloudBees deployment and configuration-as-code practices.

- Automated CloudBees Operations Center, controllers, agents, plugins, and role-based access control.
- Developed automation with Python, Groovy, Chef, Terraform, and shell scripting.
- Maintained SpecFlow/Playwright test automation and Snyk security automation.
- Supported business and DevOps teams across build, test, deployment, and platform operations.

Technology: CloudBees/Jenkins, Snyk, Chef, Terraform, Artifactory, Venafi, Docker, Kubernetes, VMware vCenter, Playwright, SpecFlow, Python, Groovy, Bash, Linux

Vector — Senior Automation Engineer

Apr 2022–Jul 2022 · Auckland, New Zealand

Automated infrastructure and deployment for a GE Advanced Distribution Management System supporting SCADA operations.

- Built reusable automation with Ansible, Packer, Terraform, and shell scripts.
- Supported deployments across Windows and AWS Outposts environments.
- Worked with SCADA and GE engineering teams to diagnose and resolve delivery issues.

Technology: AWS Outposts, VPC, Route 53, ALB, Ansible, Packer, Terraform, Bash, Linux, Windows

Adamson dela Cruz Consulting Limited — Director & Consulting Engineer

2015–2022 · Auckland, New Zealand

Delivered software architecture, cloud engineering, DevOps, IoT, full-stack development, and platform automation across a portfolio of client engagements.

SELECTED CLIENT ENGAGEMENTS

Intouch / Olympic Software — Azure DevOps Engineer

Jan 2022–Apr 2022

- Designed reusable Azure DevOps build and deployment templates for adoption across multiple clients and solutions.
- Created lean ARM infrastructure templates and trained delivery teams in the new workflow.

Technology: Azure, Azure AD, Azure DevOps, ARM templates, .NET Core

Code Avengers — Senior AWS Solutions Architect

Oct 2021–Apr 2022

- Defined and implemented the migration path from Elastic Beanstalk to containerised .NET Core microservices on ECS.
- Built infrastructure with AWS CDK, integrated education-provider SSO, and migrated video delivery to S3 and CloudFront.

Technology: AWS VPC, ECS, EKS, RDS, Route 53, ALB, WAF, Lambda, CloudFront, S3, AWS CDK, Docker, TypeScript, .NET Core

Harmony — AWS Support Engineer / Solutions Architect

Jul 2021–Oct 2021

- Supported production AWS operations and designed a new management VPC with peering, security, routing, and access controls.
- Designed EKS and MSK networking and automated clusters, worker nodes, and Kafka brokers with Terraform.
- Deployed Prometheus and Grafana for Kubernetes observability.

Technology: AWS VPC, EKS, ECS, MSK/Kafka, RDS, Route 53, ALB, WAF, Lambda, API Gateway, Terraform, Ansible, Kubernetes, Docker, Packer, Jenkins, Prometheus, Grafana

BestSeat360.tv — AWS Cloud Developer / Architect

Apr 2021–Jun 2021

- Designed a scalable AWS architecture and remote-service API for 5G-enabled video streaming.
- Built serverless integrations with AWS MediaLive and MediaStore and created gStreamer-based virtual streaming devices.

Technology: AWS Lambda, MediaLive, MediaStore, CloudFormation, IAM, AppSync, API Gateway, gStreamer, Docker, Python, Linux, Prometheus, Grafana

Yellow — Digital Platform Specialist

May 2019–Mar 2021

- Built Glue/PySpark data pipelines, crawlers, and JSON-to-Parquet transformations for data-lake and DynamoDB workflows.
- Delivered event-driven media conversion, HubSpot integration, Cognito federation, secured S3 content access, and GraphQL services.

Technology: AWS Glue, Athena, Lambda, S3, DynamoDB, EventBridge, Step Functions, MediaConvert, Cognito, KMS, AppSync, Elasticsearch, CloudFormation, Python, PySpark, Pandas, Docker

IoT Ninja — Founder & CTO

Nov 2018–Present

Founded an IoT platform that helps developers, students, and makers rapidly build connected applications using reusable frontend templates and cloud services.

- Designed the platform architecture, AWS backend, device integration, gateways, and IoT network communication.
- Built cost-effective serverless services and application frameworks using TypeScript, Next.js, React, and Vue.
- Deployed a private The Things Stack / The Things Network V3 environment.

- Integrated LoRaWAN, Cat-M1/NB-IoT, cellular, and Wi-Fi devices.

Technology: AWS IoT, Lambda, Athena, ECS, Glue, CloudFormation, IAM, AppSync, API Gateway, Kubernetes, Ansible, Docker, Linux, Python, Node.js, TypeScript, Next.js, React, Vue, LoRaWAN, The Things Stack

Spark New Zealand — IoT Consultant

Jun 2017–Nov 2018

- Designed AWS and LoRaWAN solutions for IoT proofs of concept and the Connecting Farms Wi-Fi programme.
- Provisioned and diagnosed gateways and sensors, including RF verification with spectrum-analysis tools.
- Developed embedded software for MultiTech, Kerlink, Foxconn, and LoPy devices.
- Delivered a serverless lone-worker safety solution and automated LoRa test tooling in Python.

Technology: AWS IoT, Lambda, Athena, Glue, Kinesis Firehose, CloudFormation, ECS, Ansible, Docker, Packer, Python, Node.js, React, LoRaWAN, Actility ThingPark

Air New Zealand — Platform Automation Engineer

Oct 2016–Jun 2017

- Automated AWS networking, DNS, proxies, application services, Docker workflows, repositories, and CI/CD server integration.
- Supported core Bitbucket, Jenkins, and GoCD infrastructure.

Technology: AWS CloudFormation, ECS, IAM, CodeBuild, CodeDeploy, CodePipeline, Lambda, Ansible, Docker, Packer, Python, Java, Node.js, React, Jenkins, Bitbucket, GoCD

Fletcher Building Innovation Lab — Senior Full-Stack Developer

Nov 2015–Sep 2016

- Designed and delivered prototypes and production services with measurable operational value.
- Built a Mesos/Marathon container platform, roofing geometry and material-estimation APIs, a C++ CGAL-to-Node.js interface, and ERP pricing integration.

Technology: Apache Mesos, Marathon, Docker, Node.js, C++, CGAL, Python, Angular, Ionic, MongoDB, Redis, SQL Server, Azure, Linux

AIMIA — Senior Developer

Aug 2015–Nov 2015

- Performed penetration testing and remediated CSRF, session fixation, username enumeration, cookie, caching, and content-security-policy vulnerabilities across multiple websites.

Technology: OWASP ZAP, WebScarab, Wireshark, Fiddler, CSP, PGP, X.509, SSL/TLS

EARLIER CAREER

Spark Connect — Lead Developer

Aug 2014–Aug 2015 · Auckland

Led technical implementation and operations for nationwide Wi-Fi services. Built an operations portal and Linux monitoring daemon, improved hosted-platform stability, automated VMware maintenance, and containerised core services as groundwork for SDN/NFV.

Technology: Python, Django, Node.js, AngularJS, MongoDB, Nginx, HAProxy, FreeRADIUS, Docker, Salt, VMware, Cisco and Ruckus wireless/network infrastructure

Spark Ventures — Senior Software Developer

Dec 2013–Aug 2014 · Auckland

Developed and supported core Wi-Fi services, a decoupled captive-portal API, voucher-based hosting-partner solutions, access-point provisioning, reporting, and data archiving.

Technology: ASP.NET MVC, Web API, C#, AngularJS, SQL Server, OAuth, JWT, Redis, Node.js, MongoDB, VMware, Cisco and Ruckus infrastructure

TR Group — Analyst Programmer

Mar 2013–Dec 2013 · Auckland

Technical lead for a GIS and telematics SaaS platform. Built service-oriented data access, geofencing, and fleet downtime monitoring capabilities.

Technology: ASP.NET MVC, Web API, WCF, AngularJS, SQL Server, NHibernate, Nokia HERE Maps, MapQuest

Process Intelligence — Contract Senior .NET Developer

Jul 2012–Mar 2013 · Auckland

Delivered responsive insurance software, modernised an ASP.NET portal, and integrated payment and collection workflows with MYOB.

Technology: C#, ASP.NET, Kendo UI, WinForms, SQL Server, Reporting Services, LINQ

AIA Insurance — Contract Senior .NET Developer

Jun 2011–Jul 2012 · Auckland

Designed and delivered an Application Workbench that automated insurance assessment through policy issue, plus a document-generation and communication service.

Technology: C#, ASP.NET, WCF, LINQ, Silverlight, WPF, SQL Server, IIS

Kiwiplan — Contract Senior .NET Developer

Jul 2010–Jun 2011 · Auckland

Re-engineered legacy components toward dependency injection, generated repetitive code, moved order processing to background agents, and modernised schema-driven reporting.

Technology: VB.NET, VB6, C#, WinForms, ASP.NET, WCF, SQL Server, Java, NUnit, Rhino Mocks

Imarda — Contract Senior Developer / Lead Developer

Aug 2008–Apr 2011 · Perth and Auckland

Led development of the i360 telematics platform, modernised legacy ASP/VB components, introduced rich web clients and JSON services, and created generators that greatly reduced middle-tier and backend development effort.

Technology: C#, ASP.NET, WCF, SQL Server, JavaScript, Ext JS/Sencha, jQuery, Google Maps, IIS

Connector Systems — Senior Software Engineer

Oct 2007–Aug 2008 · Auckland

Built Java asset-management and workflow-visualisation features, including SVG-based mapping and generated process diagrams.

Technology: Java, Spring MVC, Hibernate, SVG, EJB, JSP/JSTL, MySQL, Oracle

Peace Software — Senior Software Engineer

Apr 2007–Oct 2007 · Auckland

Developed arrears and pending-disconnection capabilities for an energy-sector customer and billing platform and automated builds with Ant.

Technology: Java, EJB, applets, JSP/JSTL, JUnit, MySQL, Oracle, Ant

Global Care Solutions — Senior Software Developer

Oct 2005–Apr 2007 · Bangkok, Thailand

Designed and developed an electronic medication administration record system that translated physician orders into safe administration schedules using barcode and tablet workflows.

Technology: Microsoft .NET, C#, healthcare workflow systems, barcode and tablet integration

Additional Software Engineering Roles

1999–2005 · Philippines

- **Equicom — Contract Senior Software Developer (2005):** online purchase of government-service vouchers using Microsoft .NET.
- **Devco — Contract Senior Software Developer (2004–2005):** secure Jabber-based messaging and document exchange using PKI.
- **Misys — Contract Senior Software Developer (2004):** Java/C++ trade-finance development across Linux, Solaris, Windows, Oracle, and SQL Server.
- **Michelin — Senior Software Developer (2003–2004):** modernised a supply-chain forecasting system from ASP to Java Struts/JSP.
- **Deltek Systems — Software Developer (2001–2003):** accounting-product development in VB6 and later VB.NET R&D.
- **CF Sharp — Software Developer (2000–2001):** recruitment and payroll systems for shipping and nursing businesses.
- **Informatics — Software Developer & Lecturer (1999–2000):** software development and technical instruction.

CERTIFICATIONS

Amazon Web Services

- AWS Certified Solutions Architect – Professional
- AWS Certified Solutions Architect – Associate
- AWS Certified SysOps Administrator – Associate
- AWS Certified Developer

Microsoft

- Analyzing Requirements and Defining Solution Architectures
- Designing and Implementing Web Applications with Microsoft Visual C#
- Developing XML Web Services and Server Components with Microsoft Visual C# .NET
- Developing Desktop Applications Using Microsoft Visual Basic 6.0
- Developing Distributed Applications Using Microsoft Visual Basic 6.0
- Designing and Implementing Databases with Microsoft SQL Server 2000

Additional

- The Things Network — LoRaWAN Academy
- Sun Certified Java Associate
- Meralco Foundation — Basic Electronics, Industrial Electrician, and Electro-Pneumatics

EDUCATION

- **Cisco Networking 1 & 2**, Manukau Institute of Technology, 2008
- **Computer Science**, STI College Biñan, 1997–1999
- **B.S. Marine Transportation (through third year)**, Philippine Merchant Marine Academy, 1992–1995

ADDITIONAL INTERESTS

Embedded systems, electronics, radio technologies, amateur radio (ZL1JWA), carpentry, gardening, basketball, chess, fishing, and camping. Active with family in volunteer ministry work.